

“Today, battles are fought more on the network”



Tata Power SED supplies state-of-the-art products and systems to Ministry of Defence, government of India. It's also in the process of signing agreements with a number of reputed foreign companies for becoming their offset partner. S. Gnyana Sundar as programme director-offsets, avionics and simulators is leading this effort and has signed and successfully delivered offset contracts with Israel Aerospace Industries. The latest initiative taken up by him is to build electronic manufacturing services (EMS) as an independent vertical in Tata Power SED.

Sundar speaks to Uma Gupta, an executive editor at EFY, on the state of defence electronics today and the role of Tata Power SED

■ **S. GNYANA SUNDAR**
PROGRAMME DIRECTOR-OFFSETS, AVIONICS AND SIMULATORS,
TATA POWER SED

How advanced is defence electronics today?

Technology is advancing at a furious pace. You take it from the point of view of processors, software, real-time operating systems and even communications. As technology advances, defence is trying to exploit the same by going in for most advanced techniques. Defence is becoming more and more software-controlled. Today, the battle is fought not on the battlefield but more on the network itself. We call it as network-centric warfare. For example, if you destabilise the enemy's communication network, you have won half the war. That's what we call electronic warfare. Here communication could be enemy's voice communication or their radars which can detect your aircrafts.

What new technological developments will drive further

innovation and radically change the landscape of defence electronics?

The accent is on network-centric warfare, which involves secured data links, very powerful software algorithms and advanced communication devices like software-defined radios. Unmanned remotely controlled aerial vehicles used not only for surveillance but also for weapon delivery are becoming order of the day. You are seeing that happening. USA is able to hit precisely at identified targets from drones that are remotely controlled.

Network-centric warfare involves secured data links, very powerful software algorithms and advanced communication devices like software-defined radios

Is India equipped to do so?

India has the intellectual capability to develop such systems. We are taking a balanced approach of developing systems in-house as well sourcing technology from abroad in order to